

Increased left-ventricular pressure

Biomechanical State: Regulated Variables σ_i

Homeostatic Checks: Regulated Variables - Set Points = Stimuli $\Delta\sigma_i$

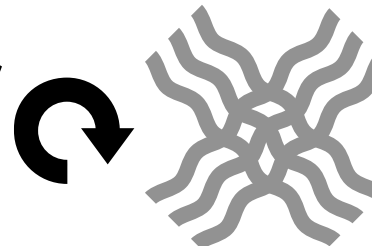
Elastin



Homeostatic stretch

Collagen

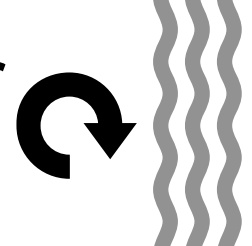
Turnover rate



Homeostatic stretch, reorientation

Sarcomeres

Turnover rate



Homeostatic stretch, reorientation

Adapted Heart Geometry, Composition, Properties